

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

**Perihalan Produk:** Mercury(II) nitrate hydrate  
**Product Description:** Mercury(II) nitrate hydrate  
**Cat No. :** 14497  
**Synonyms** Nitric acid, mercury(2+) salt, monohydrate; Mercuric nitrate monohydrate.  
**CAS No** 7783-34-8  
**Molecular Formula** Hg N2 O6 . H2 O

**Relevant identified uses of the substance or mixture and uses advised against**

**Recommended Use** Laboratory chemicals.  
**Uses advised against** No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

**E-mail address** Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                      |                   |
|------------------------------------------------------|-------------------|
| Acute oral toxicity                                  | Category 2 (H300) |
| Acute dermal toxicity                                | Category 1 (H310) |
| Acute Inhalation Toxicity - Dusts and Mists          | Category 2 (H330) |
| Specific target organ toxicity - (repeated exposure) | Category 2 (H373) |
| Acute aquatic toxicity                               | Category 1 (H400) |
| Chronic aquatic toxicity                             | Category 1 (H410) |

**Label Elements**


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## Signal Word

**Danger**

### Hazard Statements

H373 - May cause damage to organs through prolonged or repeated exposure  
H410 - Very toxic to aquatic life with long lasting effects  
H300 + H310 + H330 - Fatal if swallowed, in contact with skin or if inhaled

### Precautionary Statements

#### Prevention

P262 - Do not get in eyes, on skin, or on clothing  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P270 - Do not eat, drink or smoke when using this product  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection  
P284 - Wear respiratory protection

#### Response

P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P310 - Immediately call a POISON CENTER or doctor  
P330 - Rinse mouth  
P302 + P352 - IF ON SKIN: Wash with plenty of soap and water  
P361 + P364 - Take off immediately all contaminated clothing and wash it before reuse

#### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed  
P405 - Store locked up

#### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

### Other Hazards

Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                         | CAS No     | Weight % |
|-----------------------------------|------------|----------|
| Mercury (II) nitrate, monohydrate | 7783-34-8  | >95      |
| Mercuric nitrate                  | 10045-94-0 | -        |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### General Advice

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

#### Eye Contact

In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

#### Ingestion

Do NOT induce vomiting. Call a physician or poison control center immediately.

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**Inhalation** Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Self-Protection of the First Aider** Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

**Most important symptoms and effects, both acute and delayed**

None reasonably foreseeable.

**Indication of any immediate medical attention and special treatment needed**

**Notes to Physician** Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

**Extinguishing media**

**Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

**Extinguishing media which must not be used for safety reasons**

No information available.

**Special hazards arising from the substance or mixture**

Very toxic. Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition. Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Nitrogen oxides (NO<sub>x</sub>), Mercury oxide.

**Advice for fire-fighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

**Personal Precautions, Protective Equipment and Emergency Procedures**

Ensure adequate ventilation. Avoid dust formation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.

**Environmental precautions**

Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. Should not be released into the environment.

**Methods and Material for Containment and Cleaning Up**

Sweep up and shovel into suitable containers for disposal. Avoid dust formation.

**Reference to Other Sections**

Refer to protective measures listed in Sections 8 and 13.

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## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Avoid dust formation. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe (dust, vapor, mist, gas). Do not ingest. If swallowed then seek immediate medical assistance.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from direct sunlight.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                         | Malaysia | ACGIH TLV                            | OSHA PEL                                 |
|-----------------------------------|----------|--------------------------------------|------------------------------------------|
| Mercury (II) nitrate, monohydrate |          | TWA: 0.025 mg/m <sup>3</sup><br>Skin | (Vacated) Ceiling: 0.1 mg/m <sup>3</sup> |
| Mercuric nitrate                  |          | TWA: 0.025 mg/m <sup>3</sup><br>Skin | (Vacated) Ceiling: 0.1 mg/m <sup>3</sup> |

| Component                         | European Union | The United Kingdom                                                      | Germany                                                                                                                                                          |
|-----------------------------------|----------------|-------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mercury (II) nitrate, monohydrate |                | STEL: 0.06 mg/m <sup>3</sup> 15 min<br>TWA: 0.02 mg/m <sup>3</sup> 8 hr | TWA: 0.02 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.16 mg/m <sup>3</sup><br>Haut |
| Mercuric nitrate                  |                | STEL: 0.06 mg/m <sup>3</sup> 15 min<br>TWA: 0.02 mg/m <sup>3</sup> 8 hr | TWA: 0.02 mg/m <sup>3</sup> (8 Stunden).<br>AGW - exposure factor 8<br>TWA: 0.02 mg/m <sup>3</sup> (8 Stunden). MAK<br>Höhepunkt: 0.16 mg/m <sup>3</sup><br>Haut |

### Exposure Controls

#### Engineering Measures

Use only under a chemical fume hood. Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### **Eye Protection**

Goggles

#### **Hand Protection**

Protective gloves

#### **Skin and body protection**

Long sleeved clothing

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatibility, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

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**Respiratory Protection** When workers are facing concentrations above the exposure limit they must use appropriate certified respirators  
**Recommended Filter type:** Particulates filter conforming to EN 143  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly  
When RPE is used a face piece Fit Test should be conducted

**Hygiene Measures** Handle in accordance with good industrial hygiene and safety practice

**Environmental exposure controls** Prevent product from entering drains Do not allow material to contaminate ground water system Local authorities should be advised if significant spillages cannot be contained

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                                     |                               |                                          |
|-------------------------------------|-------------------------------|------------------------------------------|
| <b>Appearance</b>                   | Off-white                     |                                          |
| <b>Physical State</b>               | Powder Solid                  |                                          |
| <b>Odor</b>                         | Slight nitric                 |                                          |
| <b>Odor Threshold</b>               | No data available             |                                          |
| <b>pH</b>                           | 3                             |                                          |
| <b>Melting Point/Range</b>          | 77 - 79 °C / 170.6 - 174.2 °F |                                          |
| <b>Softening Point</b>              | No data available             |                                          |
| <b>Boiling Point/Range</b>          | No information available      |                                          |
| <b>Flash Point</b>                  | No information available      | <b>Method -</b> No information available |
| <b>Evaporation Rate</b>             | Not applicable                | Solid                                    |
| <b>Flammability (solid,gas)</b>     | No information available      |                                          |
| <b>Explosion Limits</b>             | No data available             |                                          |
| <b>Vapor Pressure</b>               | No information available      |                                          |
| <b>Vapor Density</b>                | Not applicable                | Solid                                    |
| <b>Specific Gravity / Density</b>   | 4.39                          |                                          |
| <b>Bulk Density</b>                 | No data available             |                                          |
| <b>Water Solubility</b>             | Soluble hydrolyses            |                                          |
| <b>Solubility in other solvents</b> | No information available      |                                          |

### Partition Coefficient (n-octanol/water)

|                                  |                          |       |
|----------------------------------|--------------------------|-------|
| <b>Autoignition Temperature</b>  | No data available        |       |
| <b>Decomposition Temperature</b> | No data available        |       |
| <b>Viscosity</b>                 | Not applicable           | Solid |
| <b>Explosive Properties</b>      | No information available |       |
| <b>Oxidizing Properties</b>      | No information available |       |

|                          |                 |
|--------------------------|-----------------|
| <b>Molecular Formula</b> | Hg N2 O6 . H2 O |
| <b>Molecular Weight</b>  | 342.6           |

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## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Light sensitive.

### Possibility of Hazardous Reactions

**Hazardous Polymerization** No information available.  
**Hazardous Reactions** None under normal processing.

### Conditions to Avoid

Avoid dust formation. Incompatible products. Excess heat. Exposure to light.

### Incompatible Materials

Alcohols. Ammonia. Cyanides. Reducing Agent. Strong acids.

### Hazardous Decomposition Products

Nitrogen oxides (NOx). Mercury oxide.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

(a) acute toxicity;  
Oral Category 2  
Dermal Category 1  
Inhalation Category 2

| Component        | LD50 Oral               | LD50 Dermal | LC50 Inhalation |
|------------------|-------------------------|-------------|-----------------|
| Mercuric nitrate | LD50 = 26 mg/kg ( Rat ) | -           | -               |

(b) skin corrosion/irritation; No data available

(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;  
Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

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There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; Category 2

Target Organs Urinary Tract.

(j) aspiration hazard; Not applicable  
Solid

Other Adverse Effects The toxicological properties have not been fully investigated.

Symptoms / effects, both acute and delayed No information available.

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

**Ecotoxicity effects** Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment. Reacts with water so no ecotoxicity data for the substance is available. May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

| Component        | Freshwater Fish            | Water Flea | Freshwater Algae | Microtox |
|------------------|----------------------------|------------|------------------|----------|
| Mercuric nitrate | 0.17 mg/l (fathead minnow) |            |                  |          |

**Persistence and degradability** Product contains heavy metals. Discharge into the environment must be avoided. Special pre-treatment is necessary  
**Persistence** May persist, based on information available.  
**Degradability** Not relevant for inorganic substances, Decomposes in contact with water.  
**Degradation in sewage treatment plant** Contains substances known to be hazardous to the environment or not degradable in waste water treatment plants. Decomposes in contact with water.

**Bioaccumulative potential** May have some potential to bioaccumulate

**Mobility in soil** Hydrolyses. The product is water soluble, and may spread in water systems. Is not likely mobile in the environment due its low water solubility. Is not likely mobile in the environment. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

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## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Should not be released into the environment Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

### **Contaminated Packaging**

Dispose of this container to hazardous or special waste collection point.

### **Other Information**

Do not flush to sewer Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not let this chemical enter the environment

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

UN-No UN1625  
Hazard Class 6.1  
Subsidiary Hazard Class P  
Packing Group II  
Proper Shipping Name MERCURIC NITRATE

### Road and Rail Transport

UN-No UN1625  
Hazard Class 6.1  
Packing Group II  
Proper Shipping Name MERCURIC NITRATE

### IATA

UN-No UN1625  
Hazard Class 6.1  
Packing Group II  
Proper Shipping Name MERCURIC NITRATE

### **Special Precautions for User**

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

### **International Inventories**

X = listed

| Component                         | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Mercury (II) nitrate, monohydrate | -         | -    | -   | X     | -    | X    | X     | X    | -        |
| Mercuric nitrate                  | 233-152-3 | X    | X   | X     | -    | X    | X     | X    | KE-23127 |

### **Note**

Note 1: The concentration stated or, in the absence of such concentrations, the generic concentrations of this Regulation (Table 3.1) or the generic concentrations of Directive 1999/45/EC (Table 3.2), are the percentages by weight of the metallic element calculated with reference to the total weight of the mixture

| Component | Seveso III Directive | Seveso III Directive | Rotterdam Convention | Basel Convention |
|-----------|----------------------|----------------------|----------------------|------------------|
|-----------|----------------------|----------------------|----------------------|------------------|



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|                                   | (2012/18/EC) - Qualifying Quantities for Major Accident Notification | (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | (PIC) | (Hazardous Waste) |
|-----------------------------------|----------------------------------------------------------------------|---------------------------------------------------------------------|-------|-------------------|
| Mercury (II) nitrate, monohydrate |                                                                      |                                                                     | X     | Annex I - Y29     |
| Mercuric nitrate                  |                                                                      |                                                                     | X     | Annex I - Y29     |

## National Regulations

**Persistent Organic Pollutant**  
**Ozone Depletion Potential**

This product does not contain any known or suspected substance  
This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

26-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

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**End of Safety Data Sheet**